

B3: Atomic and Laser Physics — MT 2020 Model Solutions

Questions 1, 2 and 4

Question 1 — Central field, Thomas–Fermi potential

(a) Central field approximation and good quantum numbers

The two-body electron–electron term $\sum_{i>j} e^2/(4\pi\epsilon_0 r_{ij})$ in the Hamiltonian prevents the Schrödinger equation from separating into one-electron problems. The **central field approximation** replaces this term by an average, spherically-symmetric screening potential $S(r_i)$ acting on each electron separately:

$$\sum_{i>j} \frac{e^2}{4\pi\epsilon_0 r_{ij}} \rightarrow \sum_i S(r_i),$$

so that the full potential seen by electron i becomes a central potential

$$U(r_i) = -\frac{Ze^2}{4\pi\epsilon_0 r_i} + S(r_i).$$

The Hamiltonian is then a sum of one-electron Hamiltonians and the total wavefunction factorises into (antisymmetrised) single-particle orbitals.

Because $U(r)$ is spherically symmetric, each orbital is labelled by

$$n, \ell, m_\ell, m_s,$$

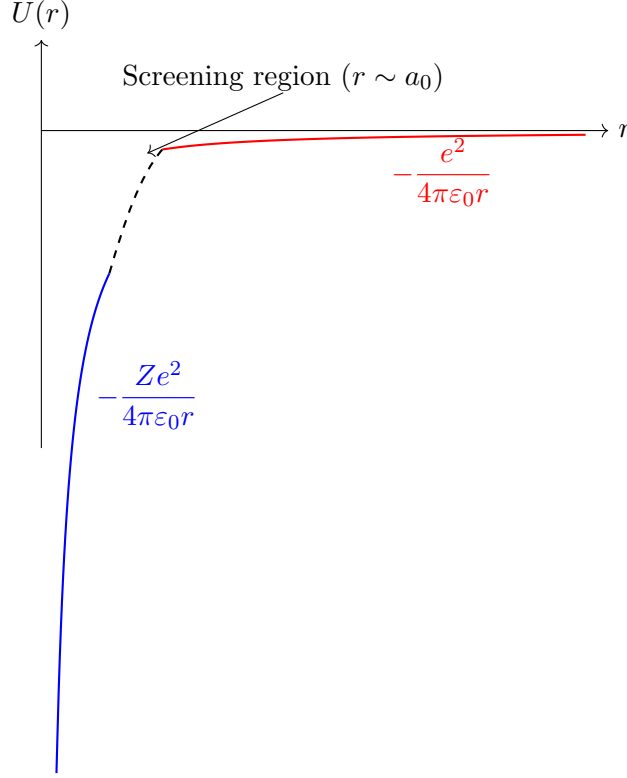
where n is the principal quantum number (unlike hydrogen, energy depends on both n and ℓ because $U(r)$ is not pure Coulombic), ℓ is the orbital angular momentum, m_ℓ its projection and $m_s = \pm\frac{1}{2}$ the spin projection. These are the *good* one-electron quantum numbers: the electrostatic repulsion term that remains after the central-field subtraction couples the individual ℓ_i and s_i so that only the total L , S (and $J = L + S$ once spin–orbit is included) are strictly good for the full atom.

(b) Sketch of the effective potential for sodium

Two limits determine the shape:

- $r \rightarrow 0$: the electron sees the bare nucleus, $U(r) \rightarrow -Ze^2/(4\pi\epsilon_0 r)$ with $Z = 11$.
- $r \rightarrow \infty$: the 10 core electrons screen all but one unit of nuclear charge, giving $U(r) \rightarrow -e^2/(4\pi\epsilon_0 r)$.

Between these two limits the screening turns on smoothly over the range of the core electrons ($\sim a_0$).



Near the origin U plunges like the bare Coulomb well of charge $Z = 11$; outside the core it flattens to the $-1/r$ tail appropriate to a single unit of net charge seen by a valence electron.

(c) Thomas–Fermi equation and length scale b

Treating the electrons as a zero-temperature Fermi gas, the number density up to momentum p is

$$n(p) = \frac{8\pi p^3}{3h^3}.$$

At each point r the local Fermi momentum $p_F(r)$ is set by the requirement that the electron is just bound, i.e. the maximum kinetic energy equals the magnitude of the local potential energy (measuring V such that $V(\infty) = 0$ and electrons have $E \leq 0$):

$$\frac{p_F^2(r)}{2m} = -eV(r) \quad \implies \quad p_F(r) = \sqrt{-2meV(r)}.$$

Hence the electron number density is

$$n_e(r) = \frac{8\pi}{3h^3} [-2meV(r)]^{3/2}. \quad (1)$$

The electrostatic potential $V(r)$ is generated by the nucleus (point charge $+Ze$) plus the electron cloud, and obeys Poisson's equation. Writing $V(r) = Z_{\text{eff}}(r) e / (4\pi\epsilon_0 r)$ with $Z_{\text{eff}}(r) = \chi(r)Z$, the nuclear singularity is accounted for by the boundary condition $\chi(0) = 1$, and away from the nucleus

$$\nabla^2 V = \frac{1}{r} \frac{d^2}{dr^2} (rV) = \frac{e n_e(r)}{\epsilon_0}.$$

Now $rV(r) = Ze\chi(r)/(4\pi\epsilon_0)$, so

$$\frac{Ze}{4\pi\epsilon_0} \frac{1}{r} \frac{d^2\chi}{dr^2} = \frac{e}{\epsilon_0} \frac{8\pi}{3h^3} \left[2me \cdot \frac{Ze\chi}{4\pi\epsilon_0 r} \right]^{3/2}. \quad (2)$$

Isolating $d^2\chi/dr^2$:

$$\frac{d^2\chi}{dr^2} = \frac{4\pi r}{Ze} \cdot \frac{8\pi}{3h^3} \left(\frac{2me^2 Z}{4\pi\epsilon_0} \right)^{3/2} \frac{\chi^{3/2}}{r^{3/2}} = C \frac{\chi^{3/2}}{r^{1/2}},$$

where

$$C = \frac{32\pi^2}{3h^3 Ze} \left(\frac{2me^2 Z}{4\pi\epsilon_0} \right)^{3/2} = \frac{32\pi^2}{3h^3} \frac{(2m)^{3/2} e^2 Z^{1/2}}{(4\pi\epsilon_0)^{3/2}}.$$

Define $r = bx$ so $d^2\chi/dr^2 = b^{-2} d^2\chi/dx^2$ and $r^{-1/2} = b^{-1/2} x^{-1/2}$. The equation becomes universal, $d^2\chi/dx^2 = \chi^{3/2}/x^{1/2}$, precisely when

$$\frac{1}{b^2} = C \frac{1}{b^{1/2}} \implies b^{3/2} = \frac{1}{C}.$$

Therefore

$$b = C^{-2/3} = \left[\frac{3h^3}{32\pi^2} \right]^{2/3} \frac{(4\pi\epsilon_0)}{(2m)e^2} Z^{-1/3}.$$

Using $h = 2\pi\hbar$ and the Bohr radius $a_0 = 4\pi\epsilon_0\hbar^2/(me^2)$, one finds

$$\boxed{b = \left(\frac{9\pi^2}{128 Z} \right)^{1/3} a_0 \approx 0.885 Z^{-1/3} a_0.} \quad (3)$$

This is the Thomas–Fermi screening length. The nuclear charge is progressively screened over this distance, and the universal function $\chi(x)$ must be solved with $\chi(0) = 1$, $\chi(\infty) = 0$.

(d) Limits of applicability

The Thomas–Fermi model treats the electrons as a continuous Fermi gas, which requires the local Fermi wavelength to be small compared with the scale over which $V(r)$ varies. This fails:

- at the *nucleus* ($r \rightarrow 0$), where $V \rightarrow -\infty$ and the density diverges unphysically;
- in the *outer reaches* of the atom, where only a few electrons are present and the continuum approximation breaks down;
- for *light atoms*, where the total electron number is too small for a statistical treatment.

It is most reliable for the core of heavy atoms (large Z).

(e) Why 4s fills before 3d in potassium

In a pure Coulomb potential ($U \propto -1/r$) the ordering is by n alone, so 3d would lie below 4s. In a screened (non-Coulombic) central field, however, orbitals with *low* ℓ have a significant probability of being found inside the core where the effective nuclear charge is much larger than its asymptotic value. This *penetration* lowers their energy. The 4s orbital, despite its larger n , has a substantial inner radial probability density (it penetrates the argon-like $1s^2 2s^2 2p^6 3s^2 3p^6$ core), whereas the 3d orbital is held away from the nucleus by its centrifugal barrier $\ell(\ell+1)\hbar^2/2mr^2$ with $\ell = 2$ and sees an almost fully screened charge $Z_{\text{eff}} \approx 1$. The increased penetration of 4s makes its energy lower than that of 3d, so the 4s level is filled first in potassium ([Ar] 4s¹).

Question 2 — Hyperfine structure in weak and strong magnetic fields

(a) Physical meaning of each term

$A \vec{I} \cdot \vec{J}$: the *magnetic hyperfine interaction*. The nuclear magnetic moment $\propto \vec{I}$ couples to the magnetic field produced at the nucleus by the electrons (and the associated contact/dipolar fields proportional to $\vec{J}$). A is the hyperfine constant.

$g_J \mu_B \vec{J} \cdot \vec{B}$: the *electronic Zeeman interaction* — the coupling of the total electronic magnetic moment $-g_J \mu_B \vec{J}/\hbar$ to the applied field $\vec{B}$. $\mu_B = e\hbar/2m_e$ is the Bohr magneton and g_J the Landé g -factor.

$-g_I \mu_N \vec{I} \cdot \vec{B}$: the *nuclear Zeeman interaction*, coupling the nuclear magnetic moment $g_I \mu_N \vec{I}/\hbar$ to $\vec{B}$. $\mu_N = e\hbar/2m_p$ is the nuclear magneton; it is $\sim m_e/m_p \approx 1/1836$ smaller than μ_B , so this term is almost always negligible compared with the electronic Zeeman term.

(b) Estimate of the weak-field threshold

The weak-field regime requires $|A \vec{I} \cdot \vec{J}| \gg |g_J \mu_B \vec{J} \cdot \vec{B}|$, i.e.

$$B \ll B^* \equiv \frac{AI}{g_J \mu_B}.$$

Taking a typical alkali ground-state hyperfine constant $A/h \sim 1$ GHz (so $A \sim h \cdot 10^9$ Hz $\approx 6.6 \times 10^{-25}$ J), $g_J \approx 2$ and $I \sim \frac{3}{2}$, and using $\mu_B = 9.27 \times 10^{-24}$ J/T,

$$B^* \sim \frac{6.6 \times 10^{-25} \cdot 1.5}{2 \cdot 9.27 \times 10^{-24}} \text{ T} \approx 0.05 \text{ T}.$$

So weak-field behaviour holds for $B \lesssim 10$ mT (of order tens of mT at most).

(c) Weak-field energy shifts

In the weak-field limit the dominant term is $A \vec{I} \cdot \vec{J}$, so $\vec{F} = \vec{I} + \vec{J}$ with $F \in \{|I - J|, \dots, I + J\}$ is a good quantum number. From $\vec{F}^2 = \vec{I}^2 + \vec{J}^2 + 2\vec{I} \cdot \vec{J}$,

$$\vec{I} \cdot \vec{J} = \frac{1}{2} [F(F + 1) - I(I + 1) - J(J + 1)] \hbar^2.$$

Absorbing the $\hbar^2$ into the definition of A ,

$$\langle A \vec{I} \cdot \vec{J} \rangle = \frac{1}{2} A K, \quad \boxed{K = F(F + 1) - I(I + 1) - J(J + 1)}.$$

The Zeeman terms are treated in first-order perturbation theory in the $|F, M_F\rangle$ basis. Using the projection (Landé) theorem, inside the $|F, M_F\rangle$ manifold

$$\langle \vec{J} \rangle = \frac{\langle \vec{F} \cdot \vec{J} \rangle}{F(F + 1) \hbar^2} \vec{F}, \quad \langle \vec{I} \rangle = \frac{\langle \vec{F} \cdot \vec{I} \rangle}{F(F + 1) \hbar^2} \vec{F},$$

with

$$\vec{F} \cdot \vec{J} = \frac{1}{2} [F(F + 1) + J(J + 1) - I(I + 1)] \hbar^2, \quad \vec{F} \cdot \vec{I} = \frac{1}{2} [F(F + 1) + I(I + 1) - J(J + 1)] \hbar^2.$$

Taking $\vec{B} = B \hat{z}$ so that $\langle F_z \rangle = M_F \hbar$, the Zeeman contribution is

$$\begin{aligned} \Delta E_Z &= g_J \mu_B B \langle J_z \rangle - g_I \mu_N B \langle I_z \rangle \\ &= \left\{ g_J \frac{F(F + 1) + J(J + 1) - I(I + 1)}{2F(F + 1)} - g_I \frac{\mu_N}{\mu_B} \frac{F(F + 1) + I(I + 1) - J(J + 1)}{2F(F + 1)} \right\} \mu_B B M_F. \end{aligned}$$

Since $\mu_N/\mu_B \sim 10^{-3}$, the nuclear term is usually dropped, giving

$$g_F = g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}$$

and the total weak-field shift is

$$\Delta E = \frac{1}{2}AK + g_F\mu_B B M_F.$$

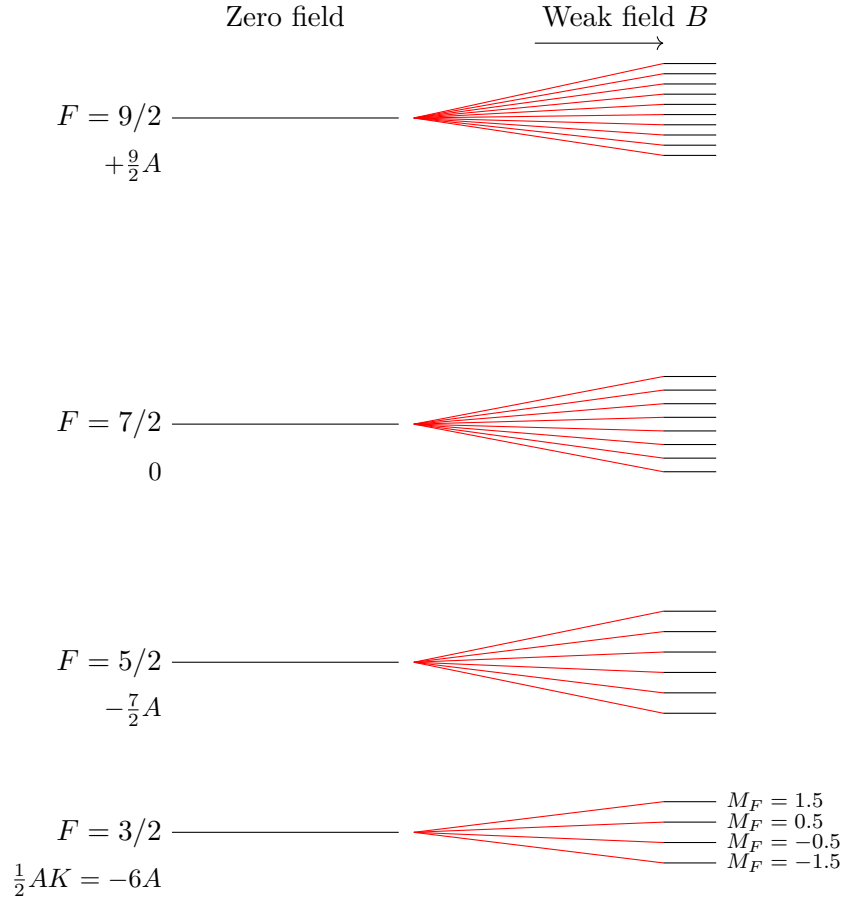
(d) Energy level diagram for $J = 3/2$, $I = 3$

Possible values: $F = |I - J|, \dots, I + J = 3/2, 5/2, 7/2, 9/2$.

Hyperfine shifts $\frac{1}{2}AK$ with $K = F(F+1) - I(I+1) - J(J+1) = F(F+1) - 12 - \frac{15}{4} = F(F+1) - \frac{63}{4}$:

F	$F(F+1)$	K
$3/2$	$15/4$	-12
$5/2$	$35/4$	-7
$7/2$	$63/4$	0
$9/2$	$99/4$	$+9$

Each level splits into $2F + 1$ equally-spaced Zeeman sublevels (4, 6, 8, 10 sublevels respectively), so the total count is $4 + 6 + 8 + 10 = 28 = (2J + 1)(2I + 1)$ as required.



Sublevel spacing within an F manifold is $g_F\mu_B B$ and differs between manifolds because g_F depends on F .

(e) Strong field for $1s^2S_{1/2}$ of hydrogen

In the strong-field regime the hyperfine term $A\vec{I} \cdot \vec{J}$ is the perturbation; J_z, I_z become good quantum numbers. For hydrogen $J = S = 1/2$, $I = 1/2$, with $g_J \rightarrow g_s$ and $g_I \rightarrow g_p$. To first order,

$$\Delta E(M_J, M_I) = g_s \mu_B B M_J - g_p \mu_N B M_I + A M_J M_I \hbar^2,$$

where we henceforth absorb $\hbar^2$ into A . The four states $|M_J, M_I\rangle$ with $M_J, M_I = \pm 1/2$ give:

(M_J, M_I)	ΔE
$(+\frac{1}{2}, +\frac{1}{2})$	$\frac{1}{2}g_s\mu_B B - \frac{1}{2}g_p\mu_N B + \frac{1}{4}A$
$(+\frac{1}{2}, -\frac{1}{2})$	$\frac{1}{2}g_s\mu_B B + \frac{1}{2}g_p\mu_N B - \frac{1}{4}A$
$(-\frac{1}{2}, +\frac{1}{2})$	$-\frac{1}{2}g_s\mu_B B - \frac{1}{2}g_p\mu_N B - \frac{1}{4}A$
$(-\frac{1}{2}, -\frac{1}{2})$	$-\frac{1}{2}g_s\mu_B B + \frac{1}{2}g_p\mu_N B + \frac{1}{4}A$

Numerator of the required ratio:

$$\begin{aligned} \Delta E_{+,-} + \Delta E_{-,-} &= \left(\frac{1}{2}g_s\mu_B B + \frac{1}{2}g_p\mu_N B - \frac{1}{4}A\right) + \left(-\frac{1}{2}g_s\mu_B B + \frac{1}{2}g_p\mu_N B + \frac{1}{4}A\right) \\ &= g_p\mu_N B. \end{aligned}$$

Denominator:

$$\begin{aligned} \Delta E_{+,-} + \Delta E_{+,+} &= \left(\frac{1}{2}g_s\mu_B B + \frac{1}{2}g_p\mu_N B - \frac{1}{4}A\right) + \left(\frac{1}{2}g_s\mu_B B - \frac{1}{2}g_p\mu_N B + \frac{1}{4}A\right) \\ &= g_s\mu_B B. \end{aligned}$$

Dividing,

$$\boxed{\frac{\Delta E_{J_z=1/2, I_z=-1/2} + \Delta E_{J_z=-1/2, I_z=-1/2}}{\Delta E_{J_z=1/2, I_z=-1/2} + \Delta E_{J_z=1/2, I_z=1/2}} = \frac{g_p\mu_N}{g_s\mu_B}}.$$

The numerator isolates the *nuclear* Zeeman coupling (electronic contributions cancel for opposite M_J) while the denominator isolates the *electronic* Zeeman coupling, yielding the ratio of the two magnetic moments.

Question 4 — Semi-classical atom–field interaction

(a) Coupled amplitude equations

Write the state as a superposition of unperturbed eigenstates:

$$\Psi(\vec{r}, t) = c_1(t) \psi_1(\vec{r}) e^{-iE_1 t/\hbar} + c_2(t) \psi_2(\vec{r}) e^{-iE_2 t/\hbar}.$$

Substituting into the TDSE $i\hbar\partial_t\Psi = (H_0 + V)\Psi$ and using $H_0\psi_n = E_n\psi_n$ leaves

$$i\hbar[\dot{c}_1\psi_1 e^{-iE_1 t/\hbar} + \dot{c}_2\psi_2 e^{-iE_2 t/\hbar}] = V(t)[c_1\psi_1 e^{-iE_1 t/\hbar} + c_2\psi_2 e^{-iE_2 t/\hbar}].$$

Project onto ψ_1^* (using orthonormality) and onto ψ_2^* :

$$i\hbar\dot{c}_1 = c_2 V_{12}(t) e^{-i\omega_0 t}, \quad i\hbar\dot{c}_2 = c_1 V_{21}(t) e^{+i\omega_0 t},$$

where $\omega_0 = (E_2 - E_1)/\hbar$ and $V_{ab}(t) = \int \psi_a^* V(t) \psi_b d\vec{r}$. For $V(t) = exE_0 \cos(\omega t)$ we have $V_{ab}(t) = E_0 \cos(\omega t) \int \psi_a^*(ex) \psi_b d\vec{r} = -E_0 \cos(\omega t) \chi_{ab}$ using the definition $\chi_{ab} = -e \int \psi_a^* x \psi_b d\vec{r}$.

Writing $\cos(\omega t) = \frac{1}{2}(e^{i\omega t} + e^{-i\omega t})$ and substituting,

$$\begin{cases} \dot{c}_1(t) = \frac{iE_0\chi_{12}}{2\hbar} [e^{i(\omega-\omega_0)t} + e^{-i(\omega+\omega_0)t}] c_2(t), \\ \dot{c}_2(t) = \frac{iE_0\chi_{12}}{2\hbar} [e^{-i(\omega-\omega_0)t} + e^{i(\omega+\omega_0)t}] c_1(t), \end{cases} \quad (4)$$

using $\chi_{21} = \chi_{12}^* = \chi_{12}$ (real for real bound-state wavefunctions and the operator x).

(b) Weak-field solution with the rotating-wave approximation

Initial conditions $c_1(0) = 1$, $c_2(0) = 0$. For a weak perturbation take $c_1(t) \approx 1$ throughout. The second equation reads

$$\dot{c}_2(t) = \frac{iE_0\chi_{12}}{2\hbar} [e^{-i(\omega-\omega_0)t} + e^{i(\omega+\omega_0)t}].$$

Near resonance $|\omega - \omega_0| \ll \omega + \omega_0$, so the second term oscillates rapidly at $\omega + \omega_0 \sim 2\omega$ and averages to zero: this is the *rotating-wave approximation* (RWA). Dropping it and integrating from 0 to t ,

$$c_2(t) = \frac{iE_0\chi_{12}}{2\hbar} \int_0^t e^{-i(\omega-\omega_0)t'} dt' = \frac{iE_0\chi_{12}}{2\hbar} \cdot \frac{e^{-i(\omega-\omega_0)t} - 1}{-i(\omega - \omega_0)}.$$

Hence

$$|c_2(t)|^2 = \frac{E_0^2\chi_{12}^2}{\hbar^2} \frac{\sin^2[(\omega - \omega_0)t/2]}{(\omega - \omega_0)^2}. \quad (5)$$

(c) Einstein B_{12} coefficient

The field is now broadband. Replace E_0^2 in (5) by its spectral content: the radiation energy density in the interval $d\omega$ is

$$dU = \frac{1}{2}\varepsilon_0 E_0^2(\omega) d\omega = u(\omega) d\omega \implies E_0^2(\omega) d\omega = \frac{2u(\omega)}{\varepsilon_0} d\omega,$$

and the upper-state probability sums incoherently over spectral components,

$$|c_2(t)|^2 = \int \frac{2\chi_{12}^2 u(\omega)}{\varepsilon_0 \hbar^2} \frac{\sin^2[(\omega - \omega_0)t/2]}{(\omega - \omega_0)^2} d\omega.$$

The sinc-squared kernel, as a function of ω , is sharply peaked at $\omega = \omega_0$ for large t . Using

$$\lim_{t \rightarrow \infty} \frac{\sin^2[(\omega - \omega_0)t/2]}{(\omega - \omega_0)^2} = \frac{\pi t}{2} \delta(\omega - \omega_0),$$

and assuming $u(\omega)$ varies slowly over the width of the kernel (i.e. broad spectrum),

$$|c_2(t)|^2 = \frac{\pi\chi_{12}^2}{\varepsilon_0 \hbar^2} u(\omega_0) t.$$

The transition rate is $d|c_2|^2/dt \equiv B_{12} u(\omega_0)/3$ once isotropic radiation is averaged (the factor $1/3$ comes from $\overline{\cos^2 \theta} = 1/3$ when averaging the dipole-field angle), giving

$$B_{12} = \frac{\pi\chi_{12}^2}{3\varepsilon_0 \hbar^2}. \quad (6)$$

(d) B_{12} for hydrogen $1s \rightarrow 2p$

We need χ_{12}^2 summed over the degenerate final $2p$ sublevels (the $2p$ manifold has $m = 0, \pm 1$).

Write $x = r \sin \theta \cos \phi$. With the tabulated spherical harmonics,

$$Y_{1,-1} - Y_{1,+1} = \sqrt{\frac{3}{8\pi}} \sin \theta (e^{-i\phi} + e^{+i\phi}) = 2\sqrt{\frac{3}{8\pi}} \sin \theta \cos \phi,$$

so

$$x = r \sin \theta \cos \phi = r \sqrt{\frac{2\pi}{3}} (Y_{1,-1} - Y_{1,+1}).$$

The matrix element decomposes as $\chi_{1s \rightarrow 2p,m} = -e I_r \cdot A_m$ with

$$I_r = \int_0^\infty R_{10}(r) r R_{21}(r) r^2 dr, \quad A_m = \sqrt{\frac{2\pi}{3}} \int Y_{0,0}^* (Y_{1,-1} - Y_{1,+1}) Y_{1,m} d\Omega.$$

Using $Y_{0,0} = 1/\sqrt{4\pi}$ and orthonormality $\int Y_{\ell,m}^* Y_{\ell',m'} d\Omega = \delta_{m',m}$, together with $Y_{\ell,-m} = (-1)^m Y_{\ell,m}^*$:

$$A_0 = 0, \quad A_{+1} = \sqrt{\frac{2\pi}{3}} \cdot \frac{1}{\sqrt{4\pi}} \cdot (-1) = -\frac{1}{\sqrt{6}}, \quad A_{-1} = \sqrt{\frac{2\pi}{3}} \cdot \frac{1}{\sqrt{4\pi}} \cdot (+1) = +\frac{1}{\sqrt{6}}.$$

Thus only $m = \pm 1$ contribute and

$$\sum_m |A_m|^2 = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}.$$

From the table: $R_{10} = 2a_0^{-3/2} e^{-r/a_0}$ and $R_{21} = (2/\sqrt{3}) a_0^{-3/2} (r/2a_0) e^{-r/2a_0} = (1/\sqrt{3}) a_0^{-5/2} r e^{-r/2a_0}$. Hence

$$\begin{aligned} I_r &= \int_0^\infty 2a_0^{-3/2} e^{-r/a_0} \cdot \frac{1}{\sqrt{3}} a_0^{-5/2} r e^{-r/2a_0} \cdot r^3 dr \\ &= \frac{2}{\sqrt{3}} a_0^{-4} \int_0^\infty r^4 e^{-3r/(2a_0)} dr \\ &= \frac{2}{\sqrt{3}} a_0^{-4} \cdot \frac{4!}{(3/2a_0)^5} = \frac{2}{\sqrt{3}} \cdot 24 \cdot \frac{2^5}{3^5} a_0 = \frac{2^8 \cdot \sqrt{3}}{3^5} a_0 = \frac{256\sqrt{3}}{243} a_0. \end{aligned}$$

(Equivalently $I_r = (2^{15/2}/3^5) a_0 \approx 1.290 a_0 \cdot \sqrt{8} \approx 3.65 a_0$.)

Summing over the three $2p$ sublevels,

$$\sum_m \chi_{1s \rightarrow 2p,m}^2 = e^2 I_r^2 \sum_m |A_m|^2 = e^2 \cdot \frac{256^2 \cdot 3}{243^2} a_0^2 \cdot \frac{1}{3} = \left(\frac{256}{243} \right)^2 \cdot e^2 a_0^2 \cdot \frac{3 \cdot 3}{3}.$$

Let me collect this carefully: $I_r^2 = (256\sqrt{3}/243)^2 a_0^2 = (256^2 \cdot 3/243^2) a_0^2$. Multiplying by $\sum_m |A_m|^2 = 1/3$:

$$\sum_m \chi_{1s,2p_m}^2 = e^2 a_0^2 \cdot \frac{256^2 \cdot 3}{243^2} \cdot \frac{1}{3} = e^2 a_0^2 \left(\frac{256}{243} \right)^2 = e^2 a_0^2 \cdot \frac{65536}{59049}.$$

Numerically $(256/243)^2 \approx 1.110$, so $\chi_{12}^2 \approx 1.110 e^2 a_0^2$ (where χ_{12}^2 denotes the sum over degenerate final states). Substituting in (6):

$$B_{12} = \frac{\pi}{3\epsilon_0 \hbar^2} \cdot \left(\frac{256}{243} \right)^2 e^2 a_0^2.$$

Inserting numerical values ($e = 1.602 \times 10^{-19} \text{ C}$, $a_0 = 5.292 \times 10^{-11} \text{ m}$, $\varepsilon_0 = 8.854 \times 10^{-12} \text{ F m}^{-1}$, $\hbar = 1.055 \times 10^{-34} \text{ J s}$):

$$\frac{e^2 a_0^2}{\varepsilon_0 \hbar^2} = \frac{(1.602 \times 10^{-19})^2 (5.292 \times 10^{-11})^2}{(8.854 \times 10^{-12})(1.055 \times 10^{-34})^2} \approx 7.29 \times 10^{20} \text{ m}^3 \text{ J}^{-1} \text{ s}^{-2},$$

giving

$$B_{12}^{1s \rightarrow 2p} = \frac{\pi}{3} \cdot \left(\frac{256}{243} \right)^2 \cdot 7.29 \times 10^{20} \text{ m}^3 \text{ J}^{-1} \text{ s}^{-2} \approx 8.5 \times 10^{20} \text{ m}^3 \text{ J}^{-1} \text{ s}^{-2}.$$

(Equivalently, expressed per unit frequency rather than per unit angular frequency, $B_{12}^{(\nu)} = 2\pi B_{12}^{(\omega)} \approx 5.3 \times 10^{21} \text{ m}^3 \text{ J}^{-1} \text{ s}^{-2}$.)